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Model Answers and Mark Schemes — First Examiner:

Paper Code: EE3.17 — Second Examiner:

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Q1 a) i) $f(i) = 1 + (i-1)K$

from the sliding window protocol

when $W > 2a+1$ an upper bound to the number of outstanding frames is $2a+1$

when $W < 2a+1$ an upper bound to the number of outstanding frames is W

Therefore

$K \sim 2a+1$ if $W > 2a+1$

$K \sim W$ if $W < 2a+1$ [5]

ii) $N_R = \sum f(i) P^{i-1}(1-P)$

$$N_R = (1-K)\sum_{i=1}^{\infty} P^{i-1}(1-P) + K\sum_{i=1}^{\infty} i P^{i-1}(1-P)$$

$$= 1 - K + \frac{K}{1-P} = \frac{1-P+KP}{1-P}$$

$$U|_{W>2a+1} = \frac{1-P}{1+2aP}$$

$$U|_{W<2a+1} = \frac{W(1-P)}{(1+2a)(1-P+WP)}$$

5

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Q4 ii) i)

MANETS

- Do not rely on fixed backbone infrastructure support.
- Dynamically changing topologies
- Mobile devices might have short battery lifetime and limited computing power
- Easy to deploy

2

Technical Issues

- Power consumption management
- Security threats
- Wireless data rate
- Scalability (memory, bandwidth, processing power)

2

Routing

- Dynamic topology (high probability of link failure)
- Every device is a potential route
- no centralised routing

2

Deployment examples

- opportunistic inter-personal communications
- Disaster relief, remote environments
- Alternative to expensive fixed infrastructure

1

ii) Intelligent vehicle

- Collision avoidance
- Driver assistance
- Collision notification

3

Specific

expected any application from above

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Q2a i) $\min T_1 = \sum x_i l_i(x_i)$

This optimisation problem minimises the total or equivalently the average delay experienced over all paths

$$T_1 = \frac{x_1^2}{3} \times x_1 + 2x_2^2 \times x_2 \qquad x_1 + x_2 = 1$$

$$\frac{3x_1^2}{3} = 6x_2^2 \qquad \text{opt condition}$$

$$x_1 = \sqrt{6}\, x_2 \qquad x_1 + x_2 = 1$$

ii) $\min T_2 = \sum l_i(x_i)$

In this case each source choose minimum delay end-to-end path. That is all path with non zero flow have the same delay. There is no regard as to the delay of the other path on the system

$$\frac{x_1^2}{3} = 2x_2^2$$

$$x_1^2 = 6x_2^2$$

$$x_1 = \sqrt{6}\, x_2 \qquad x_1 + x_2 = 1$$

$x_1^* = 0.7101$

$x_2^* = 0.2898$

iii) Suggestion

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Q2b) P1: marking of packets allow routers to distinguish amog packets from different traffic classes
- classification allow routers to treat different classes accordingly
- A service level agreement is necessary — 2

P2: provide a degree of protection amog traffic flows — 0.5
- A policing mechanism needs to be in place — 2

P3: While providing isolation amog flows it is desirable to use resources as efficiently as possible — 0.5
- Flows allow to use another flow's unused bandwidth — 2

P4: As sufficient resources will not be always available, before accepting a new call flows declare their QoS requirements — 0.5
- Call admission process is needed: network may deny service if it cannot meet needs. — 2

0.5

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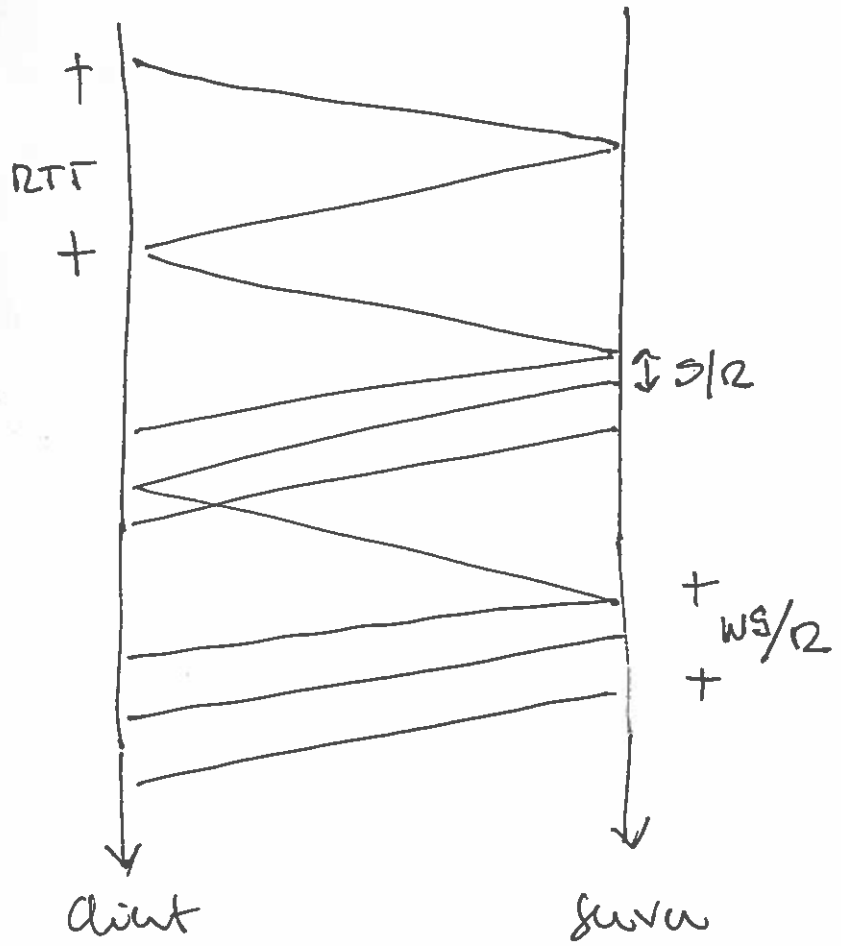
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(Q3 a) i) The TCP congestion control algorithm has three major components:

- Additive - increase, multiplicative decrease
- slow start
- reaction to time out events

+ Discussions and highlight of implementation issues 4

ii) For the case $WS/R < RTT + S/R$



$$\text{Latency} = 2RTT + O/R + (K-1)\left[\frac{S}{R} + RTT - \frac{WS}{R}\right] \quad 4$$

$$K = O/WS \quad 2$$

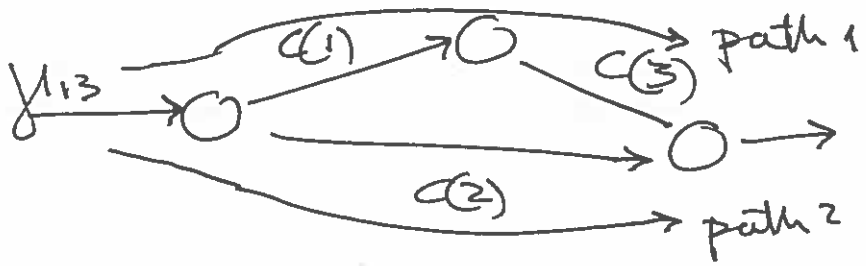
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Q3b)



i) All traffic is carried by path 2. Optimality condition:

$$\frac{C_1}{(C_1-\phi)^2} + \frac{C_3}{(C_3-\phi)^2} = \frac{C_2}{(C_2-\gamma)^2}$$

$$\frac{1}{C_1} + \frac{1}{C_3} = \frac{C_2}{(C_2-\gamma)^2} = R$$

$$C_3 + C_1 = RC_1C_3$$

$$C_1 - RC_1C_3 = -C_3$$

$$C_1(1 - RC_3) = -C_3 \longrightarrow \boxed{\frac{C_3}{RC_3-1} = C_1}$$

$$\boxed{RC_3 - 1 > 0}$$

5

ii) $C_2 = 2\gamma \qquad C_3 = 2C_2 = 4\gamma$

$$C_1 = \frac{4\gamma}{\frac{8\gamma^2}{(2\gamma-\gamma)^2} - 1} = \frac{4\gamma}{8-1} = \frac{4}{7}\gamma$$

5

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Q4 a) i) This Index is independent of the shared resource. The Index always lies between 0 and 1. For example if the Index is 0.1 means that is unfair to 90% of the users.

worst case $\frac{1}{n}$ best case 1

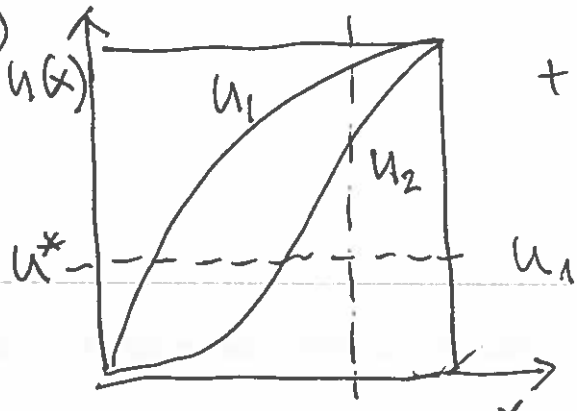
$\frac{k}{n}$ when k users equally share the resource and the other n-k users receive zero allocation. 3

ii) max-min fairness:

max-min fairness has been achieved if the allocation is feasible and an attempt to increase the allocation of flow will decrease the allocation of some other flow with an equal or smaller allocation.

- The allocation is done in infinitesimal increments to all flows until one is satisfied then among the remainder of the flows and so on until all flows are satisfied or the bandwidth is exhausted. 3

iii)



+ Discussion 3

$U_1 = U_2$ (utility max-min fair) 3

$x_1 = x_2$ (bandwidth max-min fair) 3

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(a) i) Edge Router:
- Per flow (aggregate) traffic management
- Mark packets as in-profile and out-profile

profile: pre-negotiated rate r and bucket size b (in the context of a token bucket mechanism).

Use of marking:
- class based marking
- Intra-class marking

4

ii) Core router:
- Per class traffic management
- Queueing and scheduling based on marking at the edge
- Preference given to in-profile over out-of-profile packets.
- Individual router's behaviour are referred as per-hop behaviour (PHB)
- PHB result in a different measurable forwarding performance behaviour
- PHB does not specify what mechanisms to use to ensure required PHB.
- Example of PHB:
  - Expedite forwarding
  - Assured forwarding
  - Best effort

4